

are as good as an engineer can successfully leverage it. As an analogy, if you give me the most expensive camera, it is not going to make me Ansell Adams (the American photographer and environmentalist)!"

Don't postpone design optimisation review for a later day. The design tools these days are pretty advanced and have a lot of optimisation features at various stages during the development flow. However, the timing when this design optimisation process is carried out is also critical. Performing design optimisation at an early stage and including resource utilisation as an important design consideration throughout the design flow leads to the

effect on resource utilisation. "We have options like resource sharing, controlling the fanout of signals, defining the way the tool would infer state-machines, moving logic into embedded memory, mapping memory to look-up tables, controlling design hierarchy, and so on. This helps an engineer fine-tune the design with respect to the resource he needs and those available in a target device," he adds. The synthesis tools also have automated optimisation options, which can reduce resource utilisation by up to five per cent.

Role of simulation tools in reducing resource consumption

Simulation tools form an integral part of frugal engineering as they can help reduce prototyping and testing, thereby reducing both time and costs in product development. While we cannot do away with prototyping, advancements in simulation tools can help reduce prototyping to a large extent. These tools allow simulating a real-world environment, using techniques like multiphysics modelling on a computer instead of actually fabricating and testing components. This allows not only for faster and cost-efficient experimentation but also dramatically increases the range of scenarios over which tests are performed.

While design engineers are exploring the architectures for their design, a good simulation tool provides a quick and easy way of verifying the functionality. The simulation tools also support scripting and batch modes so that the engineers can execute multiple iterations with quick turn-around times. Simulation enables developers to deal with complexity at a level where it is still easy to understand what is going on, says Genard. He explains, "Simulation has been in use since decades. From nuclear, chemical, mechanics and recently electronics, when you have something complex what you want to do is, cut in small pieces and define a higher level of abstraction. This last point always results in the

Both for engineers as well as engineering managers, design optimisation and simulation tools can be of great help in putting the safety net on to prevent resource pilferage

best engineering return on investment (ROI), when compared to performing design optimisation as a distinct and separate design function.

"Compared to traditional workflows where optimisation is done incrementally and locally at later stages of design, system-level design tools enable engineers to analyse and iterate on various architectural level design options, giving them the possibility to find optimal architectures beginning with the specification stage. In the stages that follow, like, through design elaboration, system-level simulations, implementation considerations, automatic code generation for prototyping and production purposes, engineers can continuously improve design efficiency and minimise resource utilisation," confirms Mulpur.

Taking the point further, Jain states that synthesis stage has the maximum

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